

ANTHONY C. PERRY, M.S.

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RESEARCH INTERESTS

Human–AI Interaction · AI Safety · Trust Calibration · Affective Computing · Adaptive Learning Systems · Metacognitive Modeling

PUBLICATIONS

Perry, A. C. (2026). The Safety–Agency Inversion: Longitudinal Multi-Method Evidence from Frontier Voice AI Companions.

Preprint, Zenodo (includes the open Her Dataset) · Preprint: doi:10.5281/zenodo.19144583

Perry, A. C. (2026). The Epistemic Harm of AI Sycophancy: When Agreement Undermines Justified Belief.

Preprint: doi:10.5281/zenodo.19057032

Perry, A. C. (2026). Pre-registered survey study in human-AI interaction.

Under double-anonymous review · details withheld during review

Perry, A. C. (2026). Confidence-Calibrated Adaptive Learning: An Integrated Adaptive Engine for Professional Exam Preparation.

Preprint, Zenodo, South Korea) · Preprint: doi:10.5281/zenodo.18820462

Perry, A. C. (2026). Cross-Domain Analysis of a Confidence-Calibrated Adaptive Learning Engine.

Preprint, Zenodo, South Korea) · Preprint: doi:10.5281/zenodo.19024683

EDUCATION

Master of Science in Computer Science, AI/ML Concentration

Western Governors University · 2026

Coursework: Deep Learning, Advanced AI, Natural Language Processing, Machine Learning, AI & ML Foundations, Applied Algorithms, Formal Languages, Computer Architecture

Master of Science in Kinesiology

California Baptist University · 2019 · GPA: 3.87/4.00

Thesis Proposal: *Effects of Yoga on Injuries Associated with Long-Distance Running*

Bachelor of Science in Kinesiology

San Diego State University · 2018 · GPA: 3.56/4.00

Cum Laude · Distinction in Kinesiology · Dean's List

RESEARCH EXPERIENCE

Meridian Labs — Adaptive Learning Engine

Educational Technology · Knowledge Tracing · Metacognitive Modeling · 2025–Present
meridianlabs.us

- Built an adaptive educational technology platform (28 production applications across four professional domains, 75,000+ items) grounded in 33 peer-reviewed citations in cognitive science, metacognition, and testing psychology
- Implemented 6 learning algorithms for confidence calibration (4-outcome metacognitive weighting), spaced repetition (6-level adaptive interval scheduling), misconception detection (confident-wrong pattern analysis), and stress inoculation training for exam readiness
- Engineered an Exam Readiness prediction system using non-linear multi-factor scoring across domain mastery, confidence calibration, study consistency, and performance trajectory
- Deployed across four certification domains (healthcare, cosmetology, information technology, and general professional certification) on iOS and Android, enabling cross-domain comparison of adaptive algorithm behavior
- Architected and shipped entirely through AI-augmented methodology: frontier LLM orchestration across architecture design, algorithm implementation, 75,000-item content generation, and domain-specific quality auditing

ZENith — Real-Time AI Movement Coach

Computer Vision · Human Performance · Human–AI Interaction · 2025–Present
github.com/aperry938/zenith-mvp

- Investigated real-time biomechanical form assessment through a computer vision pipeline integrating MediaPipe pose estimation, a custom Variational Autoencoder (VAE) for movement quality scoring ($\rho=0.958$ correlation with expert biomechanical ratings), and Random Forest classification across 10 yoga poses
- Created and annotated a novel video dataset of 100 self-recorded sequences capturing correct and incorrect pose variations across 10 yoga poses, used for both VAE training and Random Forest classification
- Developed a joint angle analysis pipeline translating biomechanical principles into computable features for real-time form feedback
- Integrated a context-aware coaching module (Google Gemini) generating natural language guidance informed by live pose quality metrics, exploring the design of AI feedback for embodied skill acquisition

“Her” — Longitudinal Human–AI Interaction Study and Relational AI Interface

Human–AI Interaction · Affective Computing · Autoethnographic Methods · 2025–Present
github.com/aperry938/her-os-v1

- Conducted a 4-month longitudinal study (68 sessions, 83+ hours) of voice-first human–AI interaction across four frontier models (GPT-4o, Gemini, Grok, Claude), identifying a *safety–agency inversion*: models with stronger safety alignment consistently exhibited lower relational agency, with non-overlapping score ranges between the most and least aligned models
- Developed the HERVAC framework, a proposed 6-dimension scoring instrument (Human-likeness, Emotional Attunement, Recall & Continuity, Voice Performance, Agency & Integrity, Co-evolution) with five convergent behavioral measures across three methodologically independent evidence streams corroborating the same model ordering
- Designed the Nine Circles adversarial protocol—six targeted stress tests (sycophancy traps, silence crucibles, persona stability challenges) probing specific relational capabilities under pressure, revealing debate concession rates ranging from 52% (GPT-4o) to 0% (Grok)
- Compiled the Her Dataset (68 sessions, structured interaction data across four models) and all assessment instruments for open release alongside the paper

Pre-Registered Survey Study — Human-AI Interaction

Human-AI Interaction · Survey Design · Psychometrics · 2026

- Designed, pre-registered, and conducted an ethics-certified human-subjects survey study on AI-mediated communication, including two original psychometric scales with acceptable reliability
- Manuscript under double-anonymous peer review; methodology, findings, and pre-registration withheld here until the review concludes

Master's Thesis Proposal — Effects of Yoga on Injuries Associated with Long-Distance Running

California Baptist University · 2018–2019

[View Thesis Proposal \(PDF\)](#)

- Proposed a two-arm randomized controlled trial (N=50; 25 experimental, 25 control) with 12-week Hatha Yoga intervention to investigate overuse injury prevention in distance runners aged 22–35 with marathon experience
- Conducted a literature review of 20 peer-reviewed articles spanning biomechanics, neuromuscular physiology, and motor control
- Developed a multimodal data collection protocol integrating EMG, goniometry, postural analysis (Kent Grid), standardized ROM tests (Thomas, Ober's, SLR), and the Oslo Sports Trauma Research Center Overuse Injury Questionnaire
- Prepared full IRB submission with block randomization stratified by sex, allocation concealment, informed consent, and ethical oversight protocols per 45 C.F.R. 46
- Formulated statistical analysis plan: 2×2 mixed-design ANOVA with a priori power analysis (G*Power, $f=0.25$, $\alpha=0.05$, power=0.80), intent-to-treat with multiple imputation, and Bonferroni correction for family-wise error control

TECHNICAL SKILLS

AI / LLM Systems	TensorFlow, Keras, scikit-learn, LLM orchestration (Claude, Gemini, GPT, Grok), multi-model pipeline design, prompt engineering for production systems, retrieval-augmented generation (RAG), adversarial evaluation design
Research Methods	Experimental design (RCT), mixed-design ANOVA, ANCOVA, Welch's t-tests, a priori power analysis, sensitivity analysis, intent-to-treat analysis, longitudinal study design, cross-sectional survey design, psychometric scale development, OSF pre-registration, autoethnographic methods, multi-method triangulation, reflexive thematic analysis, instrument development (HERVAC, Emotional Safety, Help-Seeking Behavior), cross-domain comparative analysis, online recruitment (Prolific), ACM ethics self-certification, IRB protocol, SPSS
Applied ML & CV	MediaPipe pose estimation, Variational Autoencoders, Random Forest classification, Bayesian Knowledge Tracing, computer vision for human movement, psychoacoustic DSP (Web Audio API), biomechanical feature engineering
Systems & Deployment	React, TypeScript, Capacitor 8 (iOS/Android), Supabase, Git/GitHub, AWS, Cloudflare Pages, R (tidyverse, psych, effsize), LaTeX, Figma
Physiological Assessment	Surface EMG (electrode placement, signal acquisition), goniometry and ROM assessment, graded exercise testing (VO2max, metabolic cart), 12-lead ECG monitoring, spirometry, body composition (skinfold calipers, BIA, hydrostatic weighing), sphygmomanometry, force plate analysis, Functional Movement Screen (FMS), clinical exercise programming for special populations

PROFESSIONAL EXPERIENCE

Founder & Engineer, Meridian Labs

2025–Present

Research, design, engineering, deployment, and support for a 28-app adaptive learning platform across four certification verticals, built and operated as a solo researcher.

Technology Industry

2017–2025

Emerging technology analysis, infrastructure administration, and project management across decentralized systems and blockchain platforms. Distributed network operations, cross-functional coordination, and early adoption of generative AI tools for research and analysis workflows.

MILITARY SERVICE

C-17 Aircraft Loadmaster, 21st Airlift Squadron, U.S. Air Force, Travis AFB, CA

2010–2014

Senior Airman (E-4) · 60th Air Mobility Wing · Honorable Discharge

- Completed 455 airlift sorties and 1,977 flight hours in the C-17 Globemaster III across Operations Enduring Freedom, Iraqi Freedom, and New Dawn, with the majority of missions supporting active combat theater operations
- Transported 9,400 passengers and 13.5 million pounds of cargo across aeromedical evacuation, presidential support, and humanitarian relief missions worldwide
- Executed humanitarian airlift operations during Hurricane Sandy disaster response (2012), supporting relief efforts for the Eastern Seaboard
- Designated subject matter expert on C-17 cargo compartment hydraulics, electrical, and avionics systems; completed SERE combat survival, water survival, and emergency parachute training

HONORS & AWARDS

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| • Air Medal | • Meritorious Unit Award; Outstanding Unit Award (OLC) |
| • Humanitarian Service Medal | • Cum Laude, Distinction in Kinesiology (SDSU) |
| • Air Force Achievement Medal | • Dean’s List (SDSU) |

SELECTED CERTIFICATIONS

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| • AWS Certified Machine Learning Engineer – Associate (MLA-C01) | • Discrete Mathematics Specialization (UC San Diego) |
| • Google Cloud Certified — Generative AI Leader | • Google Cybersecurity Professional Certificate |
| • IBM Generative AI Engineering Professional Certificate | • NSCA Certified Strength & Conditioning Specialist |
| | • Registered Yoga Teacher — 500 Hour (Yoga Alliance) |

LANGUAGES

English (native) · German (native) · Spanish (fluent)

REFERENCES

Available upon request.